

Cite this: *J. Mater. Chem. A*, 2018, 6, 2643

Multifunctional NiTiO₃ nanocoating fabrication based on the dual-Kirkendall effect enabling a stable cathode/electrolyte interface for nickel-rich layered oxides†

Ming Xu,^{ab} Linfeng Fei,^b Si-Cong Zhu,^{bc} Wei Lu,^d Yanqing Lai,^{*a} Zhian Zhang,^a Chi-Hang Lam,^b and Haitao Huang^b

Surface modification is an essential step in engineering high performance nickel-rich layered (NLO) cathode materials that can withstand the increasingly harsh environment encountered in lithium-ion batteries. However, most traditional technologies suffer disadvantages such as low diffusion kinetics, processing difficulties and/or compatibility issues. Here, based on the dual-Kirkendall effect, we report an efficient, self-guided way for fabricating a multifunctional NiTiO₃ (NTO) nanocoating to stabilize the cathode/electrolyte interface for NLO. The NTO nanocoating can epitaxially and chemically bond to the surface framework of NLO particles and eliminate surface lithium residues by evolving into a Li–Ni–Ti–O mixed phase coating that is mainly composed of Li_{2–x}TiO₃. Taking full advantage of such a novel multifunctional NTO nanocoating, the structural degradation in NLO cathodes can be effectively suppressed, leading to excellent rate capability and cyclability. This method provides insights into novel surface chemistry that prevents high performance layered electrode materials from structural degradation.

Received 23rd November 2017
Accepted 9th January 2018

DOI: 10.1039/c7ta10308e

rsc.li/materials-a

1. Introduction

Lithium-ion batteries (LIBs) are undeniably one of the greatest energy storage devices in the 21st century, driven by the steadily increasing demand for portable electronic devices in a growing market.¹ However, the limited capacities of the current commercialized cathode materials of LIBs, such as LiCoO₂ (~145 mA h g^{−1})^{2–4} and LiFePO₄ (~170 mA h g^{−1}),^{5–8} present unavoidable challenges to the deployment of clean transportation technologies.⁹ In this context, success in these fields will depend on our ability to further explore cathode materials with higher capacity and longer cycle life.^{10–12} Nickel-rich layered oxides (NLO; LiNi_{1–2x}Co_xM_xO₂, M = Mn, Al) are appealing cathode candidates for advancing high-energy-density LIBs due to their very attractive practical capacities (>200 mA h g^{−1}) and low cost.^{13–19} Nevertheless, practical

implementation of NLO materials is thwarted by their capacity loss on cycling and side-reactions with the electrolyte.^{15,20} In the past decades, substantial efforts have been devoted to the study of performance degradation in NLO materials using advanced electronic and structural diagnostic techniques aiming to develop insightful new strategies for achieving high performance NLO cathodes.^{21–28} From the structural point of view, during high-voltage (~4.5 V) cycling, part of the remaining Ni²⁺ ions (0.69 Å), with a radius similar to that of lithium ions (0.76 Å), will migrate from transition metal (TM) sites (octahedral 3a) to lithium sites (octahedral 3b), leading to detrimental structure reconstruction.^{15,20,22,24–33} It has been demonstrated that the migration is progressive on the surface of NLO particles in the form of an inactive NiO-type phase and is promoted in the regions that undergo dynamic transportation of lithium ions.^{20,22,24,28,33} Moreover, undesirable air-generating lithium impurities (Li₂CO₃ and LiOH) and side reactions with the electrolyte (HF residue) bring about unstable cathode/electrolyte interphase (SEI) formation and structural degradation in NLO cathode materials, giving rise to rapid capacity and voltage decay.^{34–36} As a consequence, the aim to address these multifaceted issues has triggered hot research topics within this field, which are of great urgency to expedite the practical application of NLO cathode materials in the near future.

For a long time, continuous research on stabilizing the structure of NLO has been dedicated to solving the above-mentioned problems.^{37–57} As one of the most effective strategies

^aSchool of Metallurgy and Environment, Central South University, Changsha 410083, P. R. China. E-mail: laiyanqing@csu.edu.cn

^bDepartment of Applied Physics, The Hong Kong Polytechnic University, Kowloon, Hong Kong. E-mail: aphhuang@polyu.edu.hk

^cCollege of Science, Key Laboratory for Ferrous Metallurgy and Resources Utilization of Ministry of Education, Wuhan University of Science and Technology, Wuhan 430065, P. R. China

^dUniversity Research Facility in Materials Characterization and Device Fabrication, The Hong Kong Polytechnic University, Kowloon, Hong Kong

† Electronic supplementary information (ESI) available. See DOI: 10.1039/c7ta10308e

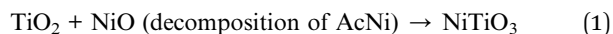
to suppress the structural degradation of NLO, nanoscale surface modification *via* coating of various layers,^{43–57} for example, polymers,^{43,44} metal oxides,^{45–47} metal fluorides,⁴⁸ or metal phosphates,⁴⁹ has been employed. Despite the significant progress in extending the cycle life of NLO cathodes by tackling lithium impurities and/or side reactions, there are still many fundamental challenges that need to be addressed. Among them, the possibility of suppressing surface reconstruction is critical, as it would guide the efficient design and fabrication of a nanocoating to mitigate performance deterioration in NLO cathode materials.^{14,20,22–24} However, progress in this area remains at an elementary level, hindered by the lack of control towards the design of an effective multifunctional coating layer that combines the advantages of bridging the material difference, stabilizing the cathode/electrolyte interface, and eliminating side-reactions.

Herein, we present a rational design of a multifunctional nanocoating based on the dual-Kirkendall effect to effectively stabilize the cathode/electrolyte interface of the NLO ($\text{LiNi}_{0.85}\text{Co}_{0.1}\text{Al}_{0.05}\text{O}_2$) cathode material. Several important design principles are adopted. First, an electrochemically stable and structurally compatible coating layer is required to bond the surface framework and stabilize the structure of NLO cathode materials. Second, the cathode/electrolyte interface should be stabilized by this coating layer that is capable of evolving into a lithium-ion conductor by eliminating surface lithium residues. Third, an epitaxial bridge between the coating layer and the surface framework of the cathode material should be engineered to effectively prevent the exfoliation of the coating layer after long-term cycling. Following the aforementioned rationales, a multifunctional NiTiO_3 (NTO) nanolayer is proposed to coat the high performance NLO cathodes. Notably, the choice of NTO is rather unique since it is one of the only a few high-performance oxides that contain nickel ions and, at the same time, exhibit excellent electrochemical stability. More importantly, after the NTO coating, we discover that the surface lithium residues evolved into a Li-Ni-Ti-O mixed phase coating that is mainly composed of $\text{Li}_{2-x}\text{TiO}_3$ (LTO). Also, we demonstrate that the NTO nanocoating can epitaxially bond to the surface framework of NLO materials by sharing the structural compatibility. As a result, the surface reconstruction of the NLO cathode in subsequent cycling can be effectively suppressed by the NTO nanocoating, bringing about a high-performance cathode material with excellent cyclability and rate capability for next-generation LIBs.

2. Results and discussion

Synthesis and fabrication of the NTO nanocoating

NTO belongs to the perovskite ABO_3 -type structure, and can be prepared *via* a typical method (Experimental section, ESI†).^{58–60} After the hydrolysis reaction of tetrabutyl titanate (TBT) with addition of $\text{NH}_3 \cdot \text{H}_2\text{O}$ and subsequent hydrothermal treatment, the prepared noncrystalline TiO_2 (TO) was well dispersed in the solution of nickel acetate tetrahydrate (AcNi). The final NTO nanoparticles were prepared by further calcination at 600 °C following the reaction below:



The X-ray diffraction pattern (Fig. S1, ESI†) demonstrates the successful synthesis of the NTO material *via* the aforementioned method. In the presence of $\text{NH}_3 \cdot \text{H}_2\text{O}$, the NTO particles can be controlled at the nanoscale, as detected by transition electron microscopy (TEM) and high resolution TEM (HRTEM) results (Fig. S2, ESI†). On the basis of the above experiments, herein, we use Kirkendall's diffusion mechanism to fabricate the multifunctional NTO nanocoating on the surface of NLO cathode materials (abbreviated as NLO-NTO).

The demonstration of net material flux by Kirkendall across the interface between two different metal specimens upon annealing was a key experiment in the development of solid-state physics and materials science (Fig. S3, ESI†).^{61–68} For the case of the pristine NLO sample, nanoscale consumption of residual lithium compounds on the surface of particles plays an important role in improving the interfacial structure, eliminating HF formation and/or reducing deleterious effects on the electrochemical performance of NLO cathodes.^{15,20,22} This fact allows us to trap the residual lithium as precursors to the coating process. A dual-Kirkendall effect was adopted to fabricate the multifunctional NTO nanocoating, making use of the bidirectional diffusion of TO species to eliminate residual lithium compounds by evolving into an inner LTO layer and to achieve an outer NTO nanocoating by reacting with NO (Fig. S4, ESI†). Based on the Kirkendall effect, the growth kinetics of the inner LTO layer and outer NTO nanocoating on the surface of NLO depend on the decomposition rate of the ionic source, the diffusion rate of ionic species, and the rate of the surface reaction.^{61,62,64,66} Indeed, the diffusivity of ionic species was experimentally and theoretically demonstrated to be governed mainly by the ionic radius,^{69,70} and in this case, $r_{\text{Ni}^{2+}}$ (0.69 Å), r_{Li^+} (0.76 Å), and $r_{\text{Ti}^{4+}}$ (0.605 Å). In this context, a suitable TO coating layer is preferentially required to ensure a uniform NTO nanocoating in the subsequent synthesis. The morphological features of the NLO-NTO sample, and the detailed structure and multifunctionality of the NTO nanocoating are systematically described in Fig. 1 and 2 and in the ESI (Fig. S5–S8†).

By following the hydrolysis of TBT and hydrothermal treatment, the NLO sample was initially coated with a uniform TO layer (schematically shown in Fig. 1a). With 1 wt% TO coating, the surfaces of the primary particles of NLO-TO microspheres were homogeneously encapsulated with the TO layer in comparison with the pristine NLO microspheres, as evidently detected by scanning electron microscopy (SEM, Fig. 1c and d) and EDX spectra (Fig. 1e). Cross-section SEM imaging (Fig. 1f) of the NLO-TO sample and the corresponding EDX line scan of the Ti element (Fig. 1g) show that, as expected, an NTO@TO core-shell structure was successfully fabricated for the NLO-TO sample, as also detected by the TEM image shown in Fig. 1h. Based on the coating strategy we proposed, such a fine core-shell structured NLO-TO sample is beneficial to the further progression of the NTO nanocoating. In contrast, beyond the 1 wt% coating amount, the increase of the TBT concentration causes an undesirable morphological change, where the TO nanoparticles agglomerated on the surface of NLO

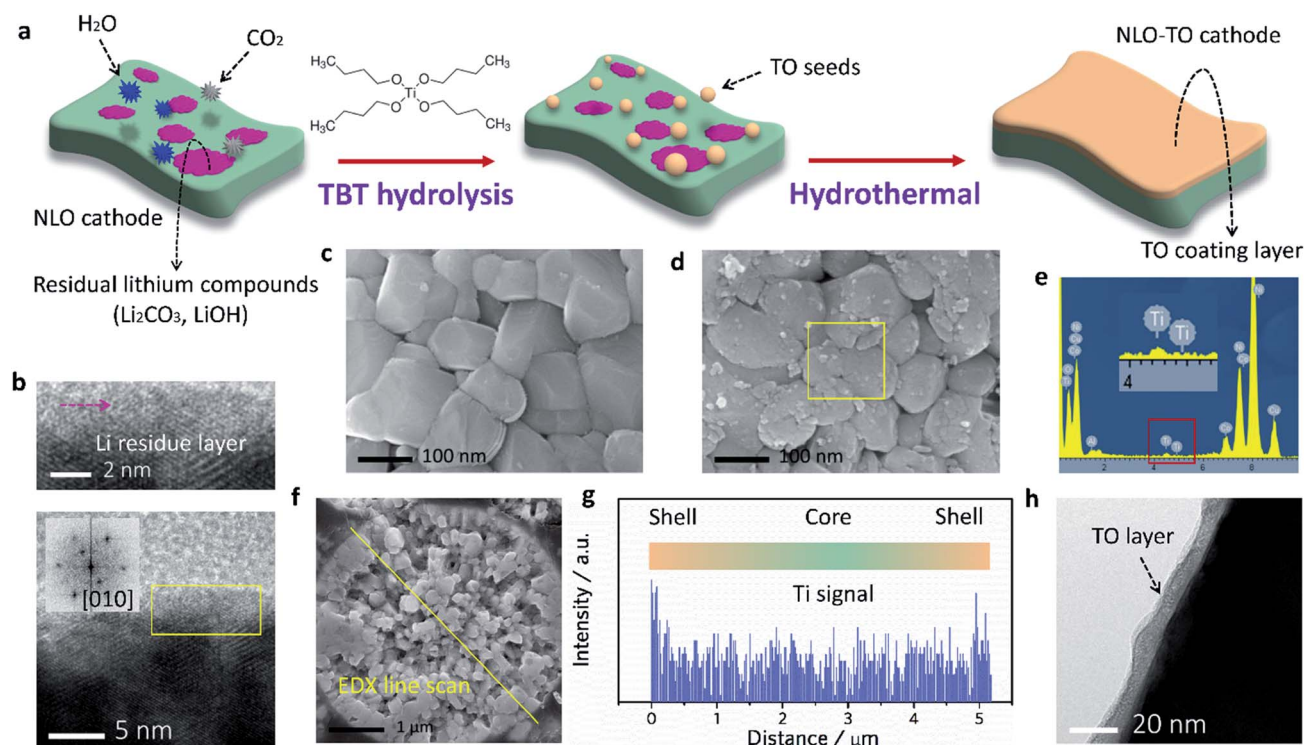


Fig. 1 Schematic view of the fabrication process of the NLO-TO sample and structural characterization of the TO coating. (a) Schematic illustration of the formation of residual lithium compounds on the surface of the pristine NLO cathode material and the fabrication process of the NLO-TO sample by a TBT hydrolysis-assisted hydrothermal method. (b) HRTEM image showing the fuzzy Li residue layer on the surface of the pristine NLO sample. SEM images of (c) the pristine NLO sample, (d) the NLO-TO sample. (e) EDX spectra of the NLO-TO sample collected from the square marked region in (d). (f) Cross-section SEM image, (g) Ti distribution along the EDX line scan shown in (f), and (h) TEM image of the NLO-TO sample.

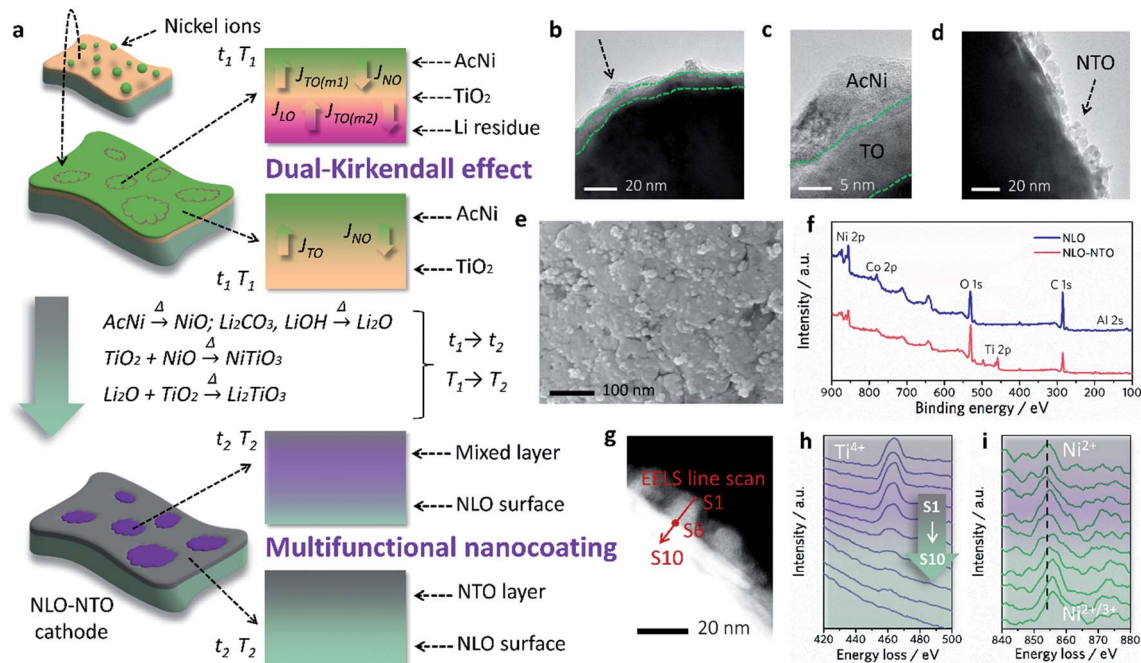


Fig. 2 (a) Schematic illustration of the fabrication process of the NLO-NTO sample based on the dual-Kirkendall effect. TEM images of (b and c) NLO-TO sample after ionic adsorption treatment. (d) TEM image, (e) SEM image, (f) XPS spectra of the NLO-NTO sample. (g) STEM image of the NLO-NTO sample and the EELS line scan of the (h) Ti-L edge, (i) Ni-L edge.

microspheres (Fig. S5 and S6, ESI†), although the addition of $\text{NH}_3 \cdot \text{H}_2\text{O}$ inhibited the hydrolysis speed of TBT. This may result in unreacted TO left in the coating layer since the amount of TO layer exceeds the required thickness for complete transformation to the NTO nanocoating.

Subsequently, the NLO-TO sample was well dispersed in AcNi solution under vigorous stirring overnight, during which the Ni^{2+} ions were homogeneously adsorbed on the surface of NLO-TO microspheres and recrystallized after water evaporation. The proposed dual-Kirkendall effect arises due to unequal diffusion rates of Ni^{2+} , Ti^{4+} and Li^+ across the boundary of two materials under thermally activated conditions (schematically shown in Fig. 2a). During the heat treatment, the growth of the inner/outer (LTO/NTO) layer includes the nucleation of Li_2O and NiO , respectively, with the former from the residual lithium compounds (LiOH , Li_2CO_3) and the latter from the thermolysis product of the AcNi layer (Fig. 2b and c), followed by the transformation from Li_2O , TO, and NiO to the LTO/NTO layer. Notably, the differences between the coating structures and morphologies of NLO-TO and NLO-NTO samples were detected by TEM (Fig. 2d) and SEM (Fig. 2e and S7, ESI†). The continuous TO nanocoating was transformed into homogeneously interconnected nanoparticles that encapsulate the NLO microspheres. As shown in Fig. 2d, voids were detected within the coating layer. The number of these voids and their formation mechanism depend on the different diffusion rates of cations or species during heat treatment, also, known as nanoscale Kirkendall porosities.^{66–68} The structures of pristine NLO, surface modified NLO-TO, and NLO-NTO particles were characterized by XRD (Fig. S8, ESI†). All materials can be indexed to a well-defined hexagonal $\alpha\text{-NaFeO}_2$ -type structure with a space group of $R\bar{3}m$.^{2–4,13,14,18} Neither impurity phases nor significant peak shifts were detected for the surface modified materials. The lattice parameters of NLO materials before and after the surface modification are almost the same (Table S2, ESI†), implying negligible surface stress induced by the NTO nanocoating on the bulk.²³ X-ray photoelectron spectroscopy (XPS) spectra have been used to determine the elements within the nanocoating, because only photoelectrons from atoms neighboring the interface within 8–10 nm are collected by XPS.⁷⁴ As shown in Fig. 2f, XPS spectra detected both Ni-2p and Ti-2p signals on the surface of the NLO-NTO sample. A better understanding of the NTO nanocoating emerges from STEM characterization, where we performed an EELS line scan across the interface between the coating layer and NLO bulk. The STEM image (Fig. 2g) of the NLO-NTO particle and the corresponding EELS line scan show the strong signals of Ti-L and Ni-L edges from the surface to its bulk (Fig. 2h and i). The diffusion of Ti ions into the NLO layer cannot be excluded during heat treatment. However, the Ti-L edge signal in the NLO layer (S10 in Fig. 2h) is negligibly weak as compared to that in the NiTiO_3 layer (S1 in Fig. 2h). Different oxidation states of the Ni element in the NTO nanocoating (Ni^{2+}) and the NLO bulk ($\text{Ni}^{2+/3+}$) were detected by the shifting of the Ni-L edge (Fig. 2i), further demonstrating the successful fabrication of the NTO nanocoating on the surface of NLO particles.

Identification and characterization of the NTO nanocoating

The success of the multifunctional NTO nanocoating relies on the dual-Kirkendall effect, where the different diffusion rates of the ionic species lead to the growth of inner LTO and outer NTO layers. The crystal growth at the solid interface follows the thermodynamic principle that is well established in previous reports.^{72,73} Taking such a mechanism into account, it was assumed that the NTO layer may chemically bond to the surface framework of NLO particles. To verify this assumption, we performed first-principles calculations using the Perdew–Burke–Ernzerhof (PBE) local density approximation (LDA) as implemented in Atomistix ToolKit.⁷⁴ (see methods in the ESI†). Considering the structural similarity of NLO and NTO (both can be indexed as a layered structure with a space group of $R\bar{3}m$),^{18,59} we would expect the formation of an epitaxial binding structure at the interface. Fig. 3a shows a special binding structure of the NTO nanocoating layer and the NLO material. The binding energy (E_b) was evaluated for the interactions between the NTO nanocoating and the bulk framework of NLO with their (003) facets facing each other. We calculated four possible configurations with different surface terminations of NTO and NLO (Fig. S9, ESI†). The most stable epitaxial and chemical binding at the interface can be achieved between the Ti-terminated NTO nanocoating and O-terminated NLO materials (Table S1, ESI†).

To verify the mixed LTO/NTO nanocoating, the atomic structure at the interface was analyzed by HRTEM (Fig. 3b). The presence of a continuous nanocoating on the surface of NLO-NTO particles is highly consistent with our observation shown in Fig. 2. A close inspection of the HRTEM image (Fig. 3c) shows an atomically sharp crystal structure, confirming the excellent crystallinity. Interestingly, the fast Fourier transform (FFT) patterns in Fig. 3d clearly illustrate the crystal structure with the (003)_{NT} and (20 $\bar{4}$)_{NT} planes of NTO along the [010]_{NT} zone. The well faceted NTO nanocoating, which is derived from the diffusion of Ti^{4+} and Ni^{2+} , remains at the outer shell of the coating layer. The thickness of the outer NTO shell is about 5–8 nm, different from the initial TO coating layer (about 12–15 nm). This may be due to the bidirectional diffusion of the TO, part of which is consumed during the formation of the inner LTO layer, consistent with the dual-Kirkendall effect. The FFT patterns shown in Fig. 3e and f illustrate the crystal structures with the (006)_{LT} and (0 $\bar{4}1$)_{LT} planes of LTO along the [100]_{LT} zone, and the (10 $\bar{3}$)_{NL} and (006)_{NL} planes of NLO along the [010]_{NL} zone confirming the existence of the inner LTO layer with a thickness of about 6 nm and the bulk NLO phase. These results are in line with our previous theoretical predictions and the simulated crystal structure (Fig. S10–S12, ESI†). For a coating layer designed to protect the NLO cathode material, it would be ideal to have the coating layer epitaxially bonded to the NLO bulk since the strong chemical bonding can provide structural stability and integrity during prolonged cycling.^{50,51} In particular, this epitaxy requires structural compatibility between the coating layer and bulk material. For the NTO nanocoating, our results show that it is highly crystallized after heat treatment and shares a similar crystal structure with the NLO bulk. Another HRTEM image (Fig. 3g) recorded near the

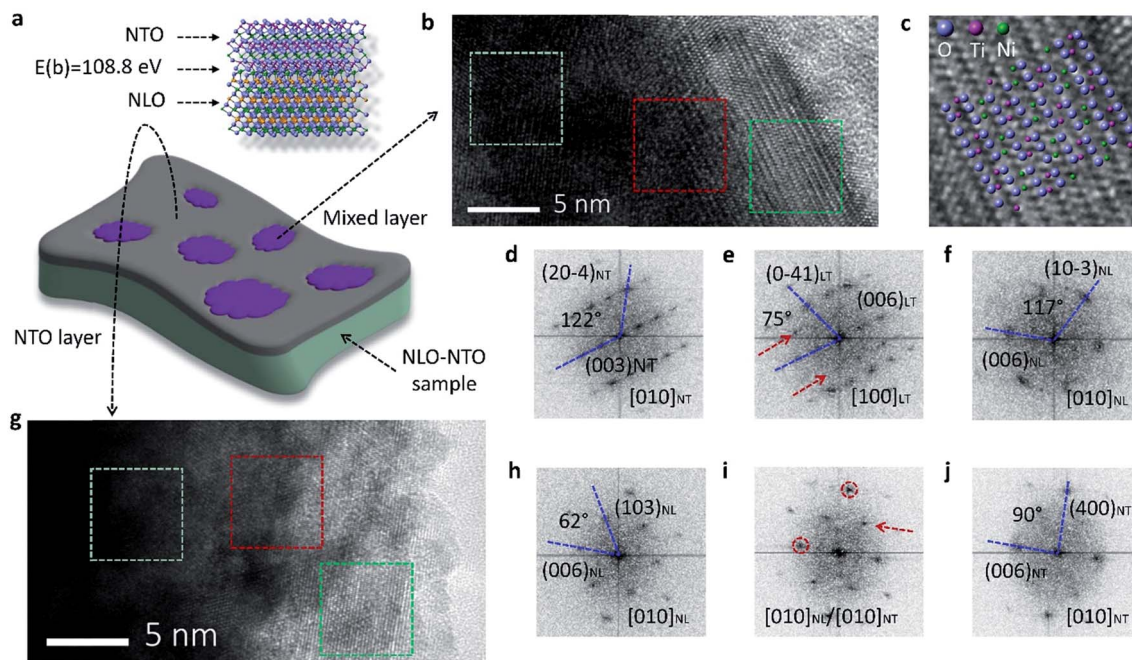


Fig. 3 (a) Schematic illustration of the multifunctional NTO nanocoating and the surface binding energy (E_b) of the NTO nanocoating on NLO calculated by DFT calculations. HRTEM images and corresponding FFT patterns of the NLO–NTO sample: (b) the interfacial structures of the mixed coating layer at the NLO–NTO surface. (c) A HRTEM image showing a detailed atomic-structure of the NTO nanocoating. (d), (e) and (f) are the FFT patterns of the green, red and cyan square marked regions in (b), respectively. The zone axes are along $[010]_{\text{NLO}}$, $[100]_{\text{LTO}}$ and $[010]_{\text{NTO}}$ for NLO, LTO and NTO, respectively. (g) The interfacial structure of the NTO nanocoating. (h), (i), and (j) are the FFT patterns of cyan, red, and green square marked regions in (g). The zone axes are along $[010]$ for both NLO and NTO.

surface of NLO–NTO particles clearly reveals the crystal structures with the $(006)_{\text{NL}}$ and $(103)_{\text{NL}}$ planes of NLO along the $[010]_{\text{NL}}$ zone axis (Fig. 3h) and the $(400)_{\text{NT}}$ and $(006)_{\text{NT}}$ planes of NTO along the $[010]_{\text{NT}}$ zone axis (Fig. 3j). Most notably, some of the FFT spots (Fig. 3i) at the NLO/NTO interface overlap with each other, confirming the epitaxial relation between the two phases (Fig. S13–S15, ESI†). When the coating amount was increased, we can detect in the thick coating layer that very few locations are able to be converted to LTO or NTO during heat treatment (Fig. S16, ESI†). These two control experiments indicate that the fabrication of a multifunctional NTO nanocoating is highly dependent on the proper control of coating amount, and the inner LTO layer or outer NTO layer can be formed when the bidirectional diffusion of the TO species proceeds at a high yield during heat treatment.

Electrochemical performance of NTO modified cathode materials

Theoretical prediction and experimental demonstration have revealed that the NTO nanocoating successfully wraps the NLO bulk, where the mixed LTO/NTO layer is generated based on the dual-Kirkendall effect. The NTO combined with the localized LTO nanocoating may act as a stabilizer for electrochemical redox reactions. The stabilized surface structure of NLO and the modified cathode/electrolyte interface may contribute to the long-term structural stability and high rate capability of NLO–NTO cathodes.^{14,15,22,24,25}

The effects of surface nanocoating on the lithiation/delithiation capacity of NLO cathodes were investigated using galvanostatic charge/discharge at 0.1C rate. The charge/discharge profiles of pristine NLO, surface modified NLO–TO and NLO–NTO samples are shown in Fig. 4a. The initial discharge capacities of NLO, NLO–TO and NLO–NTO cathodes are 216, 194, and 209 mA h g^{-1} respectively. Decreased initial capacities of surface modified NLO–TO and NLO–NTO cathodes (about 22 and 7 mA h g^{-1} , respectively) were detected. These results are in line with polarization-dependent charge transfer and redox reactions, demonstrating that the electrically insulating TO coating is the main source of capacity loss during the initial charge/discharge process. The reduced capacity loss in the NLO–NTO cathode is due to the conformal NTO layer that contributes to the suppression of polarization at the first cycle. Notably, a rather high initial coulombic efficiency (ICE) of 90.5% is observed for the NLO–NTO cathode (summarized in Table S3, ESI†). This high ICE may be due to the following three possible reasons: (i) high lithium-ion diffusion kinetics as a result of the localized LTO layer generated by consuming residual lithium compounds; (ii) suppressed surface reduction upon exposure to the electrolyte and reduced polarization at the highly delithiated state.

The rate capability is a crucial indicator for large scale applications of LIBs, such as regenerative braking and fast recharging of electric vehicles.^{1,10,12} Fig. 4b presents the rate capabilities of pristine NLO, surface modified NLO–TO and NLO–NTO cathodes with varying discharge current densities

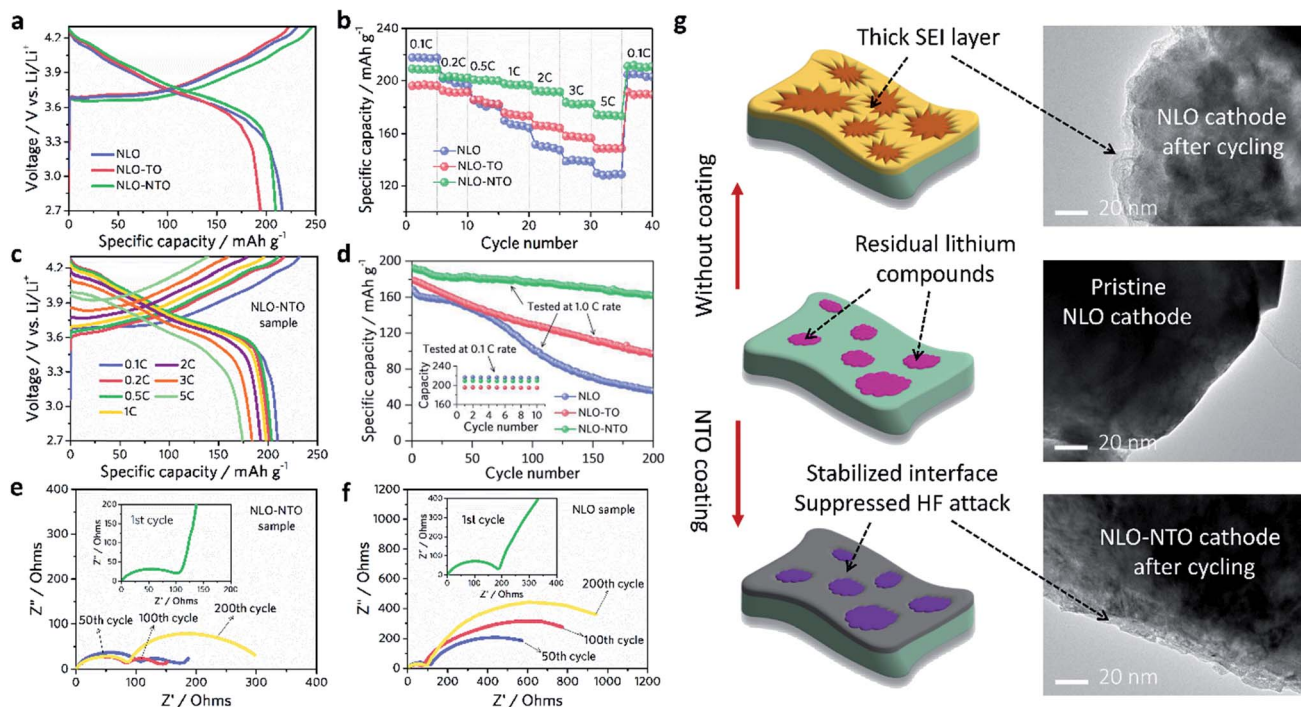


Fig. 4 Electrochemical performance of NLO, NLO-TO, and NLO-NTO cathodes: (a) initial charge/discharge profiles at 0.1C rate and (b) rate capability at different rates. Charge/discharge voltage profiles measured at different rates for (c) the NLO-NTO cathode. (d) Cycling performance at 1C rate (the inset refers to the cycling test at 0.1C rate). EIS plots of (e) NLO-NTO and (f) NLO cathodes measured at the 1st, 50th, 100th, and 200th cycles. (g) Schematics and the corresponding TEM images of the NLO and NLO-NTO cathodes after cycling.

from 0.1C to 5C at 25 °C. It is found that the NLO-NTO has the best rate capability among the three cathodes. A discharge capacity of more than 170 mA h g⁻¹ could be retained for the NLO-NTO cathode at 5C rate, corresponding to 83% capacity retention. A comprehensive summary (Table S4, ESI†) shows that this rate capability is among the best of the reported NLO cathode materials. The voltage profiles at different current rates show a lower *IR* drop in the NLO-NTO cathode (Fig. 4c) than in the pristine NLO and NLO-TO cathodes (Fig. S17, ESI†). However, when the coating amount of NTO was increased to 2.0 wt%, a low rate capability was obtained (Fig. S18, ESI†).

To evaluate the long-term cycling stability of pristine NLO, and surface modified NLO-TO and NLO-NTO cathodes, cells were tested at a current density of 1C at 25 °C, as shown in Fig. 4d. For comparison, the cycling test at 0.1C rate is also shown in the inset of Fig. 4d. As expected, the capacity (at 0.1C) delivered by the pristine NLO is the highest. The coating of NTO improves the rate performance by effectively reducing the barrier for lithium-ion transfer at the cathode/electrolyte interface and expediting the interfacial lithium-ion-transfer process.^{75,76} Hence, the capacity of NLO-NTO becomes the highest at higher rates. As expected, the NLO-NTO cathode has a very good long-term cycling stability for 200 cycles with a very low capacity decay rate of *ca.* 0.075% per cycle, suggesting that the multifunctional NTO nanocoating improves the structural stability of NLO cathodes. The NLO-TO cathode presents a relatively low capacity retention (54%), with the capacity declining from 179 to 97 mA h g⁻¹, since the TO coating is not

stable enough to protect the NLO particles against the electrolyte side reactions. Our results are in line with previous reports and the current considerations of the functional coating on cathode materials, that is, the TO layer leads to additional catalytic reactions, accelerating the loss of surface protection.⁷⁷ To evaluate the charge-transfer resistance at the cathode/electrolyte interface, electrochemical impedance spectroscopy (EIS) studies were conducted for fresh and cycled cathodes (Fig. 4e and f) and the model of the equivalent circuit is shown in Fig. S19 (ESI†). The fitting results of the representative EIS semicircles reveal the resistances of NLO cathodes due to lithium-ion migration through the surface (*R_F*) and charge-transfer at the cathode/electrolyte interface (*R_{CT}*).¹⁹ Once the lithium ions have been intercalated/deintercalated, the velocity at which they reach the surface and/or migrate through the cathode/electrolyte interface is strongly dependent on the surface structure and SEI formation of cathodes. Therefore, the charge-transfer resistance associated with the lithium ion intercalation by reduction reaction should be low in order to deliver a high capacity and rate capability. From the resistance values (Table S5, ESI†), we noticed low *R_F* values for the cycled NLO-NTO cathode, suggesting that the NTO nanocoating effectively suppressed the formation of the SEI layer. Thus, the *R_{CT}* of the NLO-NTO cathode is significantly decreased as a result of a reduced barrier for lithium-ion transfer at the cathode/electrolyte interface. By contrast, the *R_{CT}* values of the pristine NLO cathode were significantly increased after the 50th, 100th, and 200th cycles. The morphologies of NLO and NLO-

NTO cathodes after cycling (Fig. 4g) show a thick SEI layer which results from side-reactions at the cathode/electrolyte interface of the pristine NLO cathode after the repeated lithiation/delithiation processes. This result is in line with the EIS test results that the thick SEI layer inhibits the lithium-ion transfer at the cathode/electrolyte interface, resulting in high charge-transfer resistance and low ICE. By contrast, the multifunctional NTO nanocoating on the surface of the NLO–NTO cathode effectively reduces the barrier for lithium-ion migration through the cathode/electrolyte. Therefore, the NLO–NTO cathode alleviates the problems of severe electrolyte decomposition, HF attack and SEI formation upon cycling.

Previous studies have shown that the surface reconstruction of NLO materials occurs mainly during electrochemical processes,^{29,30,33} with the Ni^{2+} ions migrating from TM sites (octahedral 3a) to lithium ion sites (octahedral 3b). Modern electronic and structural diagnostic techniques have revealed that the surface reconstruction originates from dual effects involving both the cathode/electrolyte reactivity and electrochemical activation.^{21–25} From these comprehensive studies, we noticed that the HF generated from side-reactions of the electrolyte also participated in the formation of the surface reconstruction layer (containing LiF and organic components). It turns out that achieving stable high-voltage cycling of the NLO cathodes relies on structural and chemical metastability at the surfaces. These studies also clarify that the conformal

protection coating with fast lithium-ion diffusion kinetics on certain orientations of NLO is the main consideration in suppressing surface reconstruction, not HF inhibition alone.^{22,24,76,78} Therefore, by taking full advantage of the multifunctional NTO nanocoating, excellent cycling stability and rate capability can be achieved for the NLO–NTO cathode materials.

Most notably, the multifunctional NTO nanocoating combined with the localized LTO layer plays an important role in stabilizing the discharge voltage of cathodes during cycling by suppressing surface reconstruction. In comparison with the pristine NLO cathode, the NLO–NTO one has a very low voltage decay rate of ca. 0.05% per cycle (Fig. 5a). The discharge voltage profile evolution of pristine NLO and NLO–NTO cathodes upon cycling is presented in Fig. 5b and c, respectively. Without the protection coating, a poor reproducibility of voltage plateaus is obtained for the pristine NLO cathode. This is in line with the previous and our findings (Fig. 5d–f) that the surfaces of NLO particles undergo progressive reconstruction from an $R\bar{3}m$ layered structure to an $Fd\bar{3}m$ rock-salt structure originating from dual effects involving both the cathode/electrolyte reactivity and electrochemical activation, causing a voltage drop.^{15,22,24} On the contrary, with the protection of the multifunctional NTO nanocoating which epitaxially and chemically bonds to the bulk framework, the discharge voltage of NLO cathodes (Fig. 5c) is effectively stabilized and is further demonstrated by cycling tests at higher rates (Fig. S20 and S21,

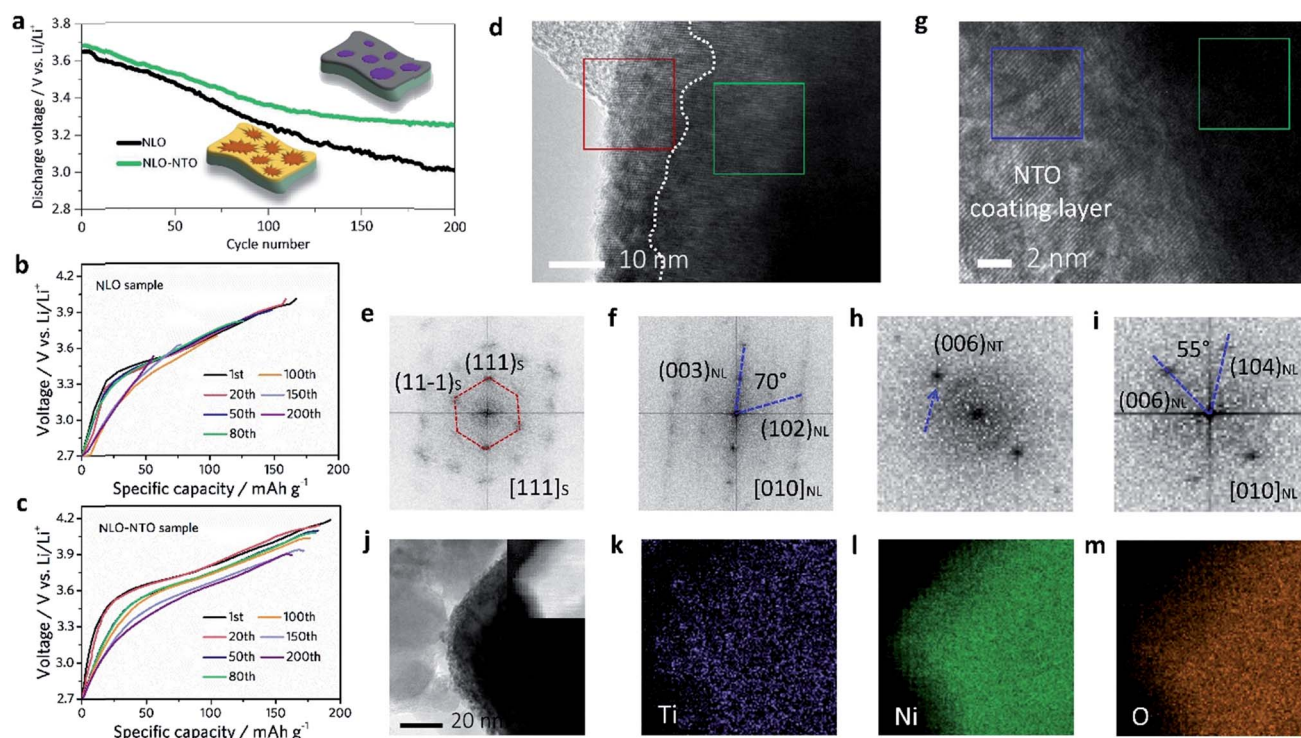


Fig. 5 (a) Discharge voltage, (b and c) discharge voltage profile evolution of NLO–NTO and NLO cathode materials upon cycling at 1.0C rate. (d) HRTEM image of the NLO cathode after cycling at 1.0C rate. The FFT patterns of (e) red and (f) green squares marked in (d) represent a spinel structure along the $[111]_s$ zone (subscript S represents a spinel structure) and a nickel-rich layered structure along the $[010]_{NL}$ zone. (g) HRTEM image of an integrated NTO coating layer on the surface of the NLO–NTO cathode after cycling at 1.0C rate. The FFT patterns of (h) blue and (i) green squares marked regions in (g) represent NTO showing a $(006)_{NT}$ facet and a NLO structure along the $[010]_{NL}$ zone. (j) TEM image (inset refers to the STEM image) and EDS mapping of (k) Ti, (l) Ni, and (m) O elements of the NLO–NTO cathode after cycling.

ESI†). As expected, a pure NLO phase is shown in the HRTEM image (Fig. 5g) and FFT patterns (Fig. 5h and i) with the (006)_{NL} and (104)_{NL} planes of NLO along the [010]_{NL} zone after the same cycling tests as for the pristine NLO. Furthermore, homogeneous distribution of Ti, Ni, and O elements in the deep cycled NLO–NTO cathode can still be detected by energy-dispersive X-ray spectroscopy (EDS) mapping (Fig. 5j–m). These findings suggest that the multifunctional NTO nanocoating plays an important role in stabilizing the surface structure of NLO materials through effective suppression of the detrimental surface reconstruction. Therefore, the NTO surface nanocoating should be more beneficial than the conventional one that paid less attention to conformity and/or chemical bonding in optimizing the structural stability for NLO cathode materials.

Surface reconstruction that originates from the dual effects involving both the cathode/electrolyte reactivity and electrochemical activation is the main unsolved problem in NLO cathode materials.^{15,22–25,76} The novel multifunctional coating layer that combines the advantages of stabilizing the surface structure and eliminating side-reactions should be taken into account in the experimental design of modified NLO cathodes.^{15,22,78} Consequently, the NTO material, a stable and conductive perovskite ABO₃-type oxide that can be prepared by an easy, fast and low cost method, becomes a material of choice for the coating layer of the NLO material. The work presented here opens new avenues for the design of a much more efficient nanoscale surface coating for suppressing the degradation in layered electrode materials that undergo structural reconstruction.

As noted previously, through careful design of a multifunctional NTO nanocoating, we have developed an efficient way based on dual-Kirkendall effect induced self-diffusion to eliminate surface lithium residue and pave a “highway” for lithium-ion migration through the cathode/electrolyte interface. In this way, we can suppress both SEI formation and side-reactions with the electrolyte on the surface of NLO cathodes. Moreover, the epitaxial interface between the NTO nanocoating and the host cathode particles can effectively prevent the exfoliation of the coating layer after long-term cycling. By reducing the barrier for lithium-ion transfer at the cathode/electrolyte interface through eliminating residual lithium compounds and suppressing surface reconstruction through epitaxial and chemical bonding of the NTO layer, we have demonstrated enhanced structural stability and rate capability for the NLO cathode during long-term cycling tests. As we have shown, the local surface structure of NLO materials can be tuned and optimized by employing this multifunctional NTO nanocoating, opening new possibilities for boosting research and applications in high-energy-density LIBs.

3. Conclusion

In this work, we have developed a facile method to fabricate a multifunctional NTO nanocoating based on a dual-Kirkendall effect. Both DFT calculations and experimental characterizations have demonstrated the chemical and epitaxial bonding interface between the NTO nanocoating and the host cathode particles. The surface reconstruction originating from the dual effects of the

cathode/electrolyte reactivity and electrochemical activation is effectively repressed by the NTO nanocoating, in addition to the unique function of eliminating surface lithium residues. Due to the advantages of such a novel multifunctional NTO nanocoating, the cycling stability and rate capability of the NLO cathode material can be greatly enhanced. A high discharge capacity of more 170 mA h g^{−1} at 5.0C rate and a very low voltage decay rate of ca. 0.05% per cycle for 200 cycles at 1.0C rate can be achieved for the NLO–NTO cathode. Thus, fabricating a multifunctional coating layer can open new avenues for the design of much more efficient nanoscale surface modification for suppressing degradation in layered electrode materials that evade structural reconstruction.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

The work was supported by the National Natural Science Foundation of China (51720105014), the Innovation Driving Plan in Central South University (2015CX001) and the Hong Kong Polytechnic University (1-ZVGH).

References

- 1 J.-M. Tarascon and M. Armand, *Nature*, 2001, **414**, 359.
- 2 T. Ohzuku and A. Ueda, *J. Electrochem. Soc.*, 1994, **141**, 2972.
- 3 Y. Shao-Horn, L. Croguennec, C. Delmas, E. C. Nelson and M. A. O'Keefe, *Nat. Mater.*, 2003, **2**, 464.
- 4 W. Cho, S. Myeong, N. Kim, S. Lee, Y. Kim, M. Kim, S. J. Kang, N. Park, P. Oh and J. Cho, *Adv. Mater.*, 2017, 1605578.
- 5 A. K. Padhi, K. S. Nanjundaswamy and J. B. Goodenough, *J. Electrochem. Soc.*, 1997, **144**, 1188.
- 6 B. Kang and G. Ceder, *Nature*, 2009, **458**, 190.
- 7 S.-Y. Chung, J. T. Bloking and Y.-M. Chiang, *Nat. Mater.*, 2002, **1**, 123.
- 8 L. Gu, C. B. Zhu, H. Li, Y. Yu, C. L. Li, S. Tsukimoto, J. Maier and Y. Ikuhara, *J. Am. Chem. Soc.*, 2011, **133**, 4661.
- 9 S. Chu, Y. Cui and N. Liu, *Nat. Mater.*, 2017, **16**, 16.
- 10 M. Armand and J.-M. Tarascon, *Nature*, 2008, **451**, 652.
- 11 H. D. Yoo, E. Markevich, G. Salitra, D. Sharon and D. Aurbach, *Mater. Today*, 2014, **17**, 110.
- 12 J. W. Choi and D. Aurbach, *Nat. Rev. Mater.*, 2016, **1**, 16013.
- 13 Y. Koyama, I. Tanaka, H. Adachi, Y. Makimura and T. Ohzuku, *J. Power Sources*, 2003, **119**, 644.
- 14 Y.-K. Sun, Z. H. Chen, H.-J. Noh, D.-J. Lee, H.-G. Jung, Y. Ren, S. Wang, C. S. Yoon, S.-T. Myung and K. Amine, *Nat. Mater.*, 2012, **11**, 942.
- 15 W. Liu, P. Oh, X. Liu, M.-J. Lee, W. Cho, S. Chae, Y. Kim and J. Cho, *Angew. Chem., Int. Ed.*, 2015, **54**, 4440.
- 16 J. H. Lee, C. S. Yoon, J.-Y. Hwang, S.-J. Kim, F. Maglia, P. Lamp, S.-T. Myung and Y.-K. Sun, *Energy Environ. Sci.*, 2016, **9**, 2152.
- 17 J. Kim, H. Cho, H. Y. Jeong, H. Ma, J. Lee, J. Hwang, M. Park and J. Cho, *Adv. Energy Mater.*, 2017, 1602559.

- 18 K. Kang, Y. S. Meng, J. Bréger, C. P. Grey and G. Ceder, *Science*, 2006, **311**, 977.
- 19 M. Jo, M. Noh, P. Oh, Y. Kim and J. Cho, *Adv. Energy Mater.*, 2014, **4**, 1301583.
- 20 N. V. Faenza, Z. W. Lebens-Higgins, P. Mukherjee, S. Sallis, N. Pereira, F. Badway, A. Halajko, G. Ceder, F. Cosandey, L. F. J. Piper and G. G. Amatucci, *Langmuir*, 2017, **33**, 9333.
- 21 J. Lu, T. P. Wu and K. Amine, *Nat. Energy*, 2017, **2**, 17011.
- 22 F. Lin, I. M. Markus, D. Nordlund, T.-C. Weng, M. D. Asta, H. L. Xin and M. M. Doeff, *Nat. Commun.*, 2014, **5**, 3529.
- 23 F. Lin, D. Nordlund, Y. Y. Li, M. K. Quan, L. Cheng, T.-C. Weng, Y. J. Liu, H. L. Xin and M. M. Doeff, *Nat. Energy*, 2016, **1**, 15004.
- 24 S.-K. Jung, H. Gwon, J. Hong, K.-Y. Park, D.-H. Seo, H. Kim, J. Hyun, W. Yang and K. Kang, *Adv. Energy Mater.*, 2014, **4**, 1300787.
- 25 P. F. Yan, J. M. Zheng, M. Gu, J. Xiao, J.-G. Zhang and C.-M. Wang, *Nat. Commun.*, 2016, **8**, 14101.
- 26 Y. Koyama, H. Arai, I. Tanaka, Y. Uchimoto and Z. Ogumi, *Chem. Mater.*, 2012, **24**, 3886.
- 27 H. J. Yu, Y. M. Qian, M. Otani, D. M. Tang, S. H. Guo, Y. B. Zhu and H. S. Zhou, *Energy Environ. Sci.*, 2014, **7**, 1068.
- 28 P. F. Yan, J. M. Zheng, J.-G. Zhang and C. M. Wang, *Nano Lett.*, 2017, **17**, 3946.
- 29 K.-K. Lee and K.-B. Kim, *J. Electrochem. Soc.*, 2000, **147**, 1709.
- 30 D. D. MacNeil, Z. Lu and J. R. Dahn, *J. Electrochem. Soc.*, 2002, **149**, A1332.
- 31 B. J. Hwang, Y. W. Tsai, D. Carlier and G. Ceder, *Chem. Mater.*, 2003, **15**, 3676.
- 32 M. G. Kim, H. J. Shin, J.-H. Kim, S.-H. Park and Y.-K. Sun, *J. Electrochem. Soc.*, 2005, **152**, A1320.
- 33 T. Sasaki, T. Nonaka, H. Oka, C. Okuda, Y. Itou, Y. Kondo, Y. Takeuchi, Y. Ukyo, K. Tatsumi and S. Muto, *J. Electrochem. Soc.*, 2009, **156**, A289.
- 34 D.-H. Cho, C.-H. Jo, W. Cho, Y.-J. Kim, H. Yashiro, Y.-K. Sun and S.-T. Myung, *J. Electrochem. Soc.*, 2014, **161**, A920.
- 35 G. V. Zhuang, G. Y. Chen, J. Shim, X. Y. Song, P. N. Ross and T. J. Richardson, *J. Power Sources*, 2004, **134**, 293.
- 36 K. Shizuka, C. Kiyohara, K. Shima and Y. Takeda, *J. Power Sources*, 2007, **166**, 233.
- 37 Y.-K. Sun, D.-H. Kim, C. S. Yoon, S.-T. Myung, J. Prakash and K. Amine, *Adv. Funct. Mater.*, 2010, **20**, 485.
- 38 H.-J. Noh, S.-T. Myung, Y. J. Lee and Y.-K. Sun, *Chem. Mater.*, 2014, **26**, 5973.
- 39 J. Xie, A. D. Sendek, E. D. Cubuk, X. K. Zhang, Z. Y. Lu, Y. J. Gong, T. Wu, F. F. Shi, W. Liu, E. J. Reed and Y. Cui, *ACS Nano*, 2017, 7019–7027.
- 40 D.-W. Jun, C. S. Yoon, U.-H. Kim and Y.-K. Sun, *Chem. Mater.*, 2017, **29**, 5048.
- 41 B.-B. Lim, S.-T. Myung, C. S. Yoon and Y.-K. Sun, *ACS Energy Lett.*, 2016, **1**, 283–289.
- 42 U.-H. Kim, S.-T. Myung, C. S. Yoon and Y.-K. Sun, *ACS Energy Lett.*, 2017, **2**, 1848–1854.
- 43 S. H. Ju, I.-S. Kang, Y.-S. Lee, W.-K. Shin, S. Kim, K. Shin and D.-W. Kim, *ACS Appl. Mater. Interfaces*, 2014, **6**, 2546.
- 44 C. V. Rao, A. L. M. Reddy, Y. Ishikawa and P. M. Ajayan, *ACS Appl. Mater. Interfaces*, 2011, **3**, 2966.
- 45 A. M. Wise, C. M. Ban, J. N. Weker, S. Misra, A. S. Cavanagh, Z. C. Wu, Z. Li, M. S. Whittingham, K. Xu, S. M. George and M. F. Toney, *Chem. Mater.*, 2015, **27**, 6146.
- 46 Y. P. Chen, Y. Zhang, B. J. Chen, Z. Y. Wang and C. Lu, *J. Power Sources*, 2014, **256**, 20.
- 47 X. H. Xiong, Z. X. Wang, G. C. Yan, H. J. Guo and X. H. Li, *J. Power Sources*, 2014, **245**, 183.
- 48 S.-U. Woo, C. S. Yoon, K. Amine, I. Belharouak and Y.-K. Sun, *J. Electrochem. Soc.*, 2007, **154**, A1005.
- 49 Y. Huang, F.-M. Jin, F.-J. Chen and L. Chen, *J. Power Sources*, 2014, **256**, 1.
- 50 M.-J. Lee, S. Lee, P. Oh, Y. Kim and J. Cho, *Nano Lett.*, 2014, **14**, 993.
- 51 Z. Z. Wu, S. P. Ji, T. C. Liu, Y. D. Duan, S. Xiao, Y. Lin and K. Xu, *Nano Lett.*, 2016, **16**, 6357.
- 52 L. J. Li, M. Xu, Q. Yao, Z. Y. Chen, L. B. Song, Z. A. Zhang, C. H. Gao, P. Wang, Z. Y. Yu and Y. Q. Lai, *ACS Appl. Mater. Interfaces*, 2016, **8**, 30879.
- 53 M.-H. Park, M. Noh, S. Lee, M. Ko, S. Chae, S. Sim, S. Choi, H. Kim, H. Nam, S. Park and J. Cho, *Nano Lett.*, 2014, **14**, 4083.
- 54 L. J. Li, Z. Y. Chen, Q. B. Zhang, M. Xu, X. Zhou, H. L. Zhu and K. L. Zhang, *J. Mater. Chem. A*, 2015, **3**, 894.
- 55 Z. Y. Wang, S. S. Huang, B. J. Chen, H. Wu and Y. Zhang, *J. Mater. Chem. A*, 2014, **2**, 19983.
- 56 H. Kim, M. G. Kim, H. Y. Jeong, H. Nam and J. Cho, *Nano Lett.*, 2015, **15**, 2111.
- 57 J. Choi, J. Kim, K.-T. Lee, J. Lim, J. Lee and Y. S. Yun, *Adv. Mater. Interfaces*, 2016, **3**, 1600784.
- 58 S. Moghiminia, H. Farsi and H. Raissi, *Electrochim. Acta*, 2014, **132**, 512.
- 59 J. J. Xu, W. Ding, W. L. Zhao, W. Zhao, Z. L. Hong and F. Q. Huang, *ACS Energy Lett.*, 2017, **2**, 659.
- 60 L. T. Yu, J. Liu, X. J. Xu, L. G. Zhang, R. Z. Hu, J. W. Liu, L. Z. Ouyang, L. C. Yang and M. Zhu, *ACS Nano*, 2017, **11**, 5120.
- 61 H. J. Fan, M. Knez, R. Scholz, K. Nielsch, E. Pippel, D. Hesse, M. Zacharias and U. Gösele, *Nat. Mater.*, 2006, **5**, 627.
- 62 R. A. Masumura, B. B. Rath and C. S. Pande, *Acta Mater.*, 2002, **50**, 4535.
- 63 E. González, J. Arbiol and V. F. Puntes, *Science*, 2011, **334**, 1377.
- 64 S. Jana, J. W. Chang and R. M. Rioux, *Nano Lett.*, 2013, **13**, 3618.
- 65 A. Paul, M. J. H. van Dal, A. A. Kodentsov and F. J. J. van Loo, *Acta Mater.*, 2004, **52**, 623.
- 66 Y. D. Yin, C. K. Erdonmez, A. Cabot, S. Hughes and A. P. Alivisatos, *Adv. Funct. Mater.*, 2006, **16**, 1389.
- 67 Y. G. Sun, X. B. Zuo, S. K. R. S. Sankaranarayanan, S. Peng, B. Narayanan and G. Kamath, *Science*, 2017, **356**, 303.
- 68 J. G. Railsback, A. C. J. Peck, J. W. Wang and J. B. Tracy, *ACS Nano*, 2010, **4**, 1913.
- 69 M. Miyamoto and H. Takeda, *Nature*, 1983, **303**, 602.
- 70 M. Morioka, *Geochim. Cosmochim. Acta*, 1981, **45**, 1573.

- 71 B. Philippe, R. Dedryvère, J. Allouche, F. Lindgren, M. Gorgoi, H. Rensmo, D. Gonbeau and K. Edström, *Chem. Mater.*, 2012, **24**, 1107.
- 72 D. S. Li, M. H. Nielsen, J. R. I. Lee, C. Frandsen, J. F. Banfield and J. J. De Yoreo, *Science*, 2012, **336**, 1014.
- 73 W.-I. Liang, X. W. Zhang, Y. L. Zan, M. Pan, C. Czarnik, K. Bustillo, J. Xu, Y.-H. Chu and H. M. Zheng, *J. Am. Chem. Soc.*, 2015, **137**, 14850.
- 74 M. Khazaei, S. U. Lee, F. Pichierri and Y. Kawazoe, *ACS Nano*, 2008, **2**, 939.
- 75 J. Lu, Q. Peng, W. Y. Wang, C. Y. Nan, L. H. Li and Y. D. Li, *J. Am. Chem. Soc.*, 2013, **135**, 1649.
- 76 W. D. Li, A. Dolocan, P. Oh, H. Celio, S. Park, J. Cho and A. Manthiram, *Nat. Commun.*, 2017, **8**, 14589.
- 77 X. F. Zhang, I. Belharouak, L. Li, Y. Lei, J. W. Elam, A. Nie, X. Q. Chen, R. S. Yassar and R. L. Axelbaum, *Adv. Energy Mater.*, 2013, **3**, 1299.
- 78 S. Kalluri, M. Yoon, M. Jo, H. K. Liu, S. X. Dou, J. Cho and Z. P. Guo, *Adv. Mater.*, 2017, 1605807.